Geometric Twins – Homework Questions

CBSE Class 7 Mathematics · Ganita Prakash (Part 2) · Chapter: Geometric Twins

Name: _____ Class/Section: _____ Date: _____

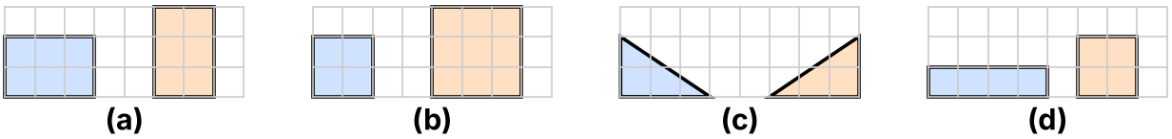
Instructions: Answer all 30 questions. Show your reasoning where needed. Figures are for reference. Equal tick marks on sides mean those sides are equal; equal arcs mean those angles are equal.

Section A – Congruent Figures

Q1. Fill in the blanks: Two figures are called **congruent** if they have the same _____ and the same _____.

Q2. Write **True** or **False**:
(a) Two circles, each of radius 4 cm, are congruent.
(b) Any two squares are always congruent.
(c) Two line segments, each of length 5 cm, are congruent.
(d) A photograph and its enlarged copy are congruent.

Q3. Look at the pairs of shapes drawn on the square grid below. Which pairs are congruent? (Hint: imagine cutting out one shape and placing it on the other. You may turn or flip it.)



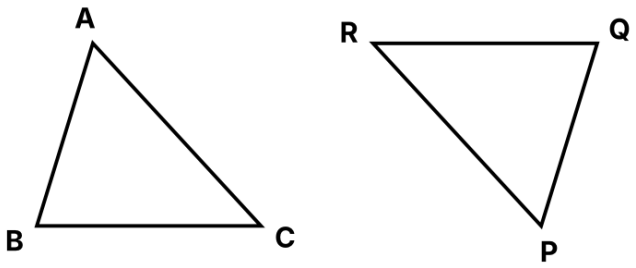
Q4. Which measurement(s) do you need to know to draw an exact copy of:
(a) a circle? (b) a square? (c) a rectangle?

Q5. Answer in Yes or No:
(a) ∠A = 50° and ∠B = 50°. Are ∠A and ∠B congruent?
(b) PQ = 3 cm and XY = 3.5 cm. Are line segments PQ and XY congruent?

Q6. Write any **two** examples of congruent objects that you see in your daily life.

Section B – Corresponding Parts

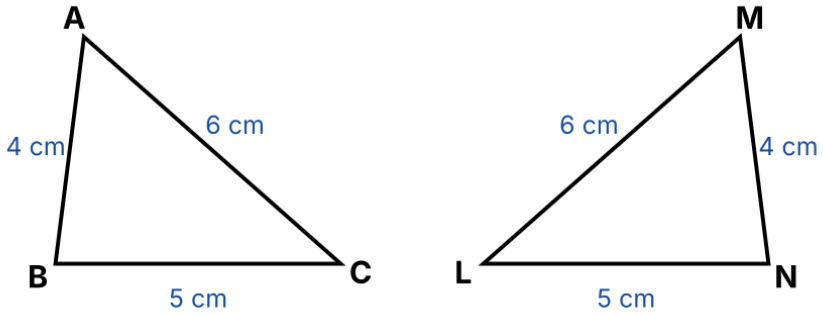
Q7. In the figure, $\Delta ABC \cong \Delta PQR$. Complete the following:
(a) AB ↔ _____ (b) BC ↔ _____ (c) ∠B ↔ _____ (d) ∠C ↔ _____



Q8. $\Delta DEF \cong \Delta LMN$. If DE = 6 cm, EF = 7 cm and DF = 5 cm, find the lengths of LM, MN and LN.

Q9. $\Delta XYZ \cong \Delta KLM$. If $\angle X = 60°$ and $\angle Y = 70°$, find ∠K, ∠L and ∠M.

Q10. Look at the two triangles below. They are congruent by matching equal sides. Complete the congruence statement in the correct order: $\Delta ABC \cong \Delta$_____

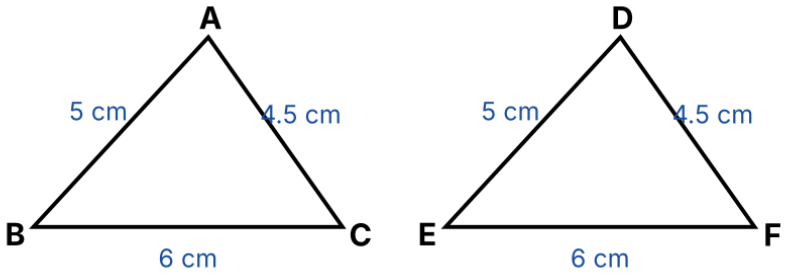


Q11. If $\Delta ABC \cong \Delta PQR$, which one of the following is **true**?
(a) AB = QR (b) ∠A = ∠P (c) BC = PR (d) ∠B = ∠R

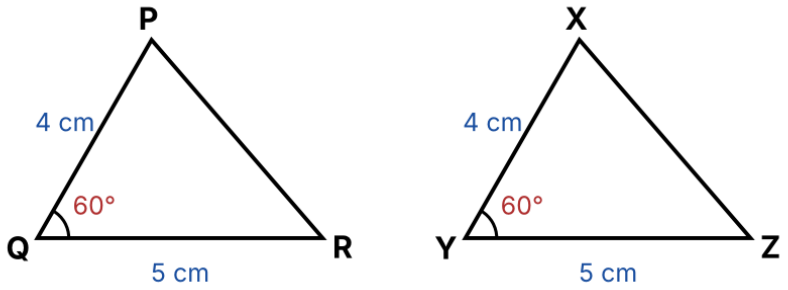
Section C – Conditions for Congruence of Triangles

Q12. Write the full form of each congruence condition:
(a) SSS (b) SAS (c) ASA (d) RHS

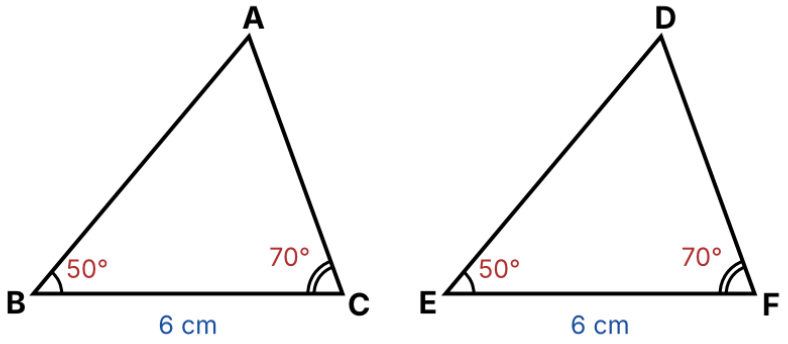
Q13. Are the two triangles below congruent? If yes, write the condition used and the congruence statement.



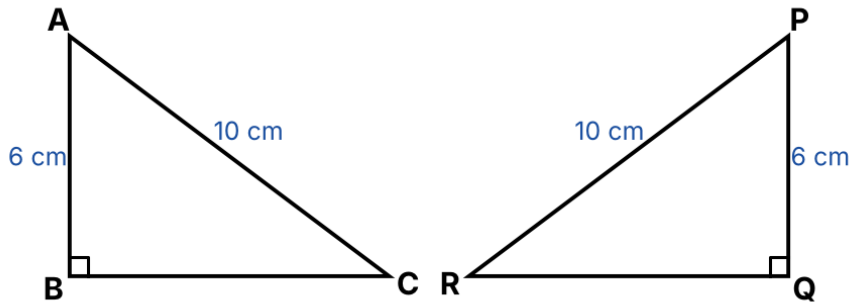
Q14. Are the two triangles below congruent? If yes, write the condition used and the congruence statement.



Q15. In the figure, ∠B = ∠E = 50°, BC = EF = 6 cm and ∠C = ∠F = 70°. Are the triangles congruent? Write the condition used and the congruence statement.



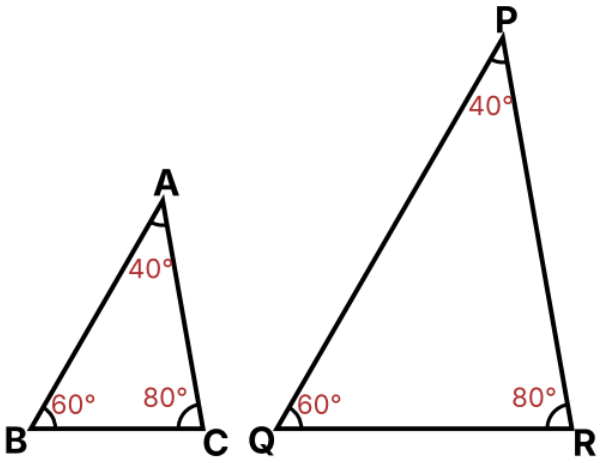
Q16. In the figure, ∠B = ∠Q = 90°, AC = PR = 10 cm and AB = PQ = 6 cm. Are the triangles congruent? Write the condition used and the congruence statement.



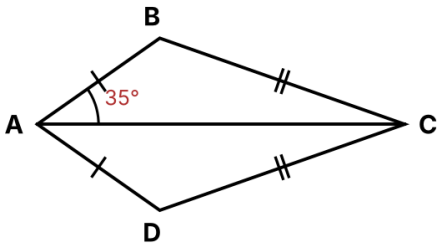
Q17. Fill in the blank: In the SAS condition, the angle must be the angle _____ the two given sides.

Q18. For ΔABC and ΔPQR, name the condition (SSS, SAS, ASA or RHS) that shows they are congruent:
(a) AB = PQ, BC = QR, CA = RP
(b) AB = PQ, ∠A = ∠P, AC = PR
(c) ∠A = ∠P, AB = PQ, ∠B = ∠Q
(d) ∠B = ∠Q = 90°, AC = PR, BC = QR

Q19. In the figure, both triangles have angles 40°, 60° and 80°. Are the two triangles congruent? Give a reason.



Q20. In the figure, AB = AD and CB = CD.
(a) Which condition shows that ΔABC ≅ ΔADC?
(b) If ∠BAC = 35°, find ∠DAC.

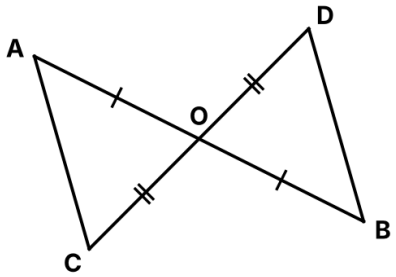


Q21. In the figure, line segments AB and CD cross each other at O, with OA = OB and OC = OD.

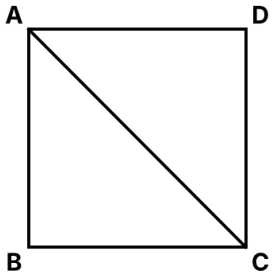
(a) Why is ∠AOC = ∠BOD?

(b) Which condition shows that ΔAOC ≅ ΔBOD?

(c) If AC = 5 cm, find BD.

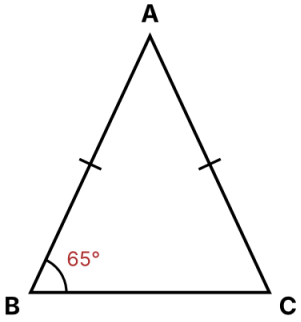


Q22. ABCD is a square and AC is joined. Is ΔABC ≅ ΔCDA? Which condition did you use?



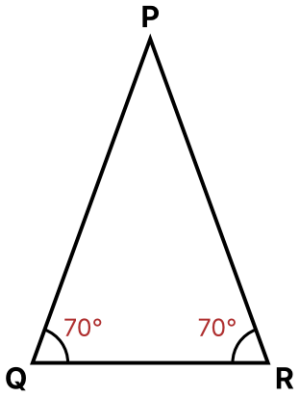
Section D – Isosceles and Equilateral Triangles

Q23. In ΔABC, AB = AC and ∠B = 65°. Find ∠C and ∠A.



Q24. What is the measure of each angle of an equilateral triangle?

Q25. In ΔPQR, ∠Q = ∠R = 70°.
(a) Which two sides of the triangle are equal? (b) Find ∠P.



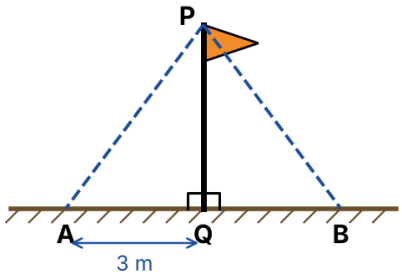
Q26. The angle between the two equal sides of an isosceles triangle is 80°. Find each of the other two angles.

Q27. Each of the two equal angles of an isosceles triangle is 45°. Find the third angle. What special name can you give to this triangle?

Q28. Can an isosceles triangle have two equal angles of 95° each? Give a reason.

Section E – Congruence Around Us

Q29. A flag pole PQ stands straight on the ground (∠PQA = ∠PQB = 90°). Two ropes PA and PB of **equal length** are tied from the top P to the points A and B on the ground.
(a) Which condition shows that ΔPQA ≅ ΔPQB?
(b) If QA = 3 m, what is QB?



Q30. Ravi draws a triangle with sides 3 cm, 4 cm and 5 cm. His friend Asha draws a triangle with sides 5 cm, 3 cm and 4 cm. Are the two triangles congruent? Give a reason.